

$$StartTx(t) \triangleq$$

$$\wedge tx_status[t] = \text{"Inactive"}$$

$$\wedge \text{LET } sess \triangleq Workload[t].session$$

$$lvl \triangleq Workload[t].level$$

$$\text{IN IF } lvl \in \{\text{"PC"}, \text{"SI"}, \text{"SER"}\}$$

$$\text{THEN } tx_snapshot' = [tx_snapshot \text{ EXCEPT } ![t] = shard_ct]$$

$$\text{ELSE IF } lvl \in \{\text{"CC"}, \text{"PSI"}\}$$

$$\text{THEN } tx_snapshot' = [tx_snapshot \text{ EXCEPT } ![t] = client_dep[session]]$$

$$\text{ELSE } RA$$

$$tx_snapshot' =$$

$$[tx_snapshot \text{ EXCEPT } ![t] = client_last_commit[session]]$$

$$\wedge tx_status' = [tx_status \text{ EXCEPT } ![t] = \text{"Active"}]$$

$$\wedge \text{UNCHANGED } \langle client_dep, client_last_commit, shard_ct, tx_checkset, \\ tx_buffer, tx_commit_time, pc, store, op_seq \rangle$$